

E-COMMERCE DATA BASE

Phase 1: Data Entry, Organization, and Cleaning

Expected Proficiency: Beginner to Intermediate

Before you can analyze data, you have to ensure it is clean and structured properly.

Step 1: Data Entry & Table Creation

Importing/Entry: If you are given a raw text file (.txt or .csv), go to the Data tab > Get & Transform Data > From Text/CSV to import it properly.

Table Creation: Select your dataset and press Ctrl + T (Windows) or Cmd + T (Mac) to turn it into an official Excel Table. This automatically applies headers, filtering arrows, and professional formatting.

Step 2: Data Cleaning & Transformation

Removing Duplicates: Highlight your data, go to the Data tab, and click Remove Duplicates. Excel will let you choose which columns to check for identical rows.

Handling Missing/Inconsistent Data: * Use Find & Replace (Ctrl + H) to fix inconsistent text (e.g., changing "TN" to "Tamil Nadu" uniformly).

To find blank cells quickly, press F5 > click Special > select Blanks. You can then fill them with 0 or "N/A" so your formulas don't break later.

Phase 2: Excel Formulas & Functions

Expected Proficiency: Intermediate

You will be graded on your ability to use specific logic and lookup functions. Here is how to use the exact ones listed in your rubric:

Basic & Text Functions

=SUM(range) and =AVERAGE(range): Used for quick totals and averages.

=TEXT(value, format_text): Converts a number or date into a specific text format. For example, =TEXT(A1, "dd-mmm-yyyy") turns a standard date into "12-Jun-2026".

Logical & Conditional Functions

=IF(logical_test, value_if_true, value_if_false): Evaluates a condition.

Example: =IF(B2>50, "Pass", "Fail")

=COUNTIF(range, criteria): Counts cells that meet a single condition.

Example: =COUNTIF(C2:C20, "Completed")

Lookup Functions

=XLOOKUP(lookup_value, lookup_array, return_array): The modern, safest way to search for data.

Example: =XLOOKUP(E2, A2:A10, B2:B10) (Looks for the value of E2 in column A, and returns the matching value from column B).

=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup]): The traditional lookup.

Remember that your lookup value must be in the very first column of your selected table array, and use FALSE at the end for an exact match.

Phase 3: Descriptive Statistics

Expected Proficiency: Intermediate

This section requires you to summarize the core tendencies of your numeric data.

Mean (Average): Use =AVERAGE(range)

Median (Middle Value): Use =MEDIAN(range)

Mode (Most Frequent Value): Use =MODE.SNGL(range)

Standard Deviation (SD): Use =STDEV.S(range) for sample data or =STDEV.P(range) if you have the entire population data.

Using the Analysis ToolPak (Highly Recommended)

Instead of typing these formulas individually, you can generate a complete statistical summary report in seconds:

Go to the Data tab.

Look for Data Analysis on the far right. (If it's not there, go to File > Options > Add-ins > Excel Add-ins "Go" > Check "Analysis ToolPak").

Select Descriptive Statistics and click OK.

Input your data range, check the box for Summary Statistics, and choose where you want the output table to appear.

Phase 4: Data Summarization & Visualization

Expected Proficiency: Intermediate

This is where you transform raw data into a visual story.

Step 1: Create a Pivot Table

Click anywhere inside your clean data table.

Go to Insert > PivotTable.

Drag your categorical data (like regions, categories, or names) into the Rows field, and your numeric data (like sales or quantities) into the Values field.

Step 2: Build Visualizations

Bar/Column Charts: Best for comparing items across categories.

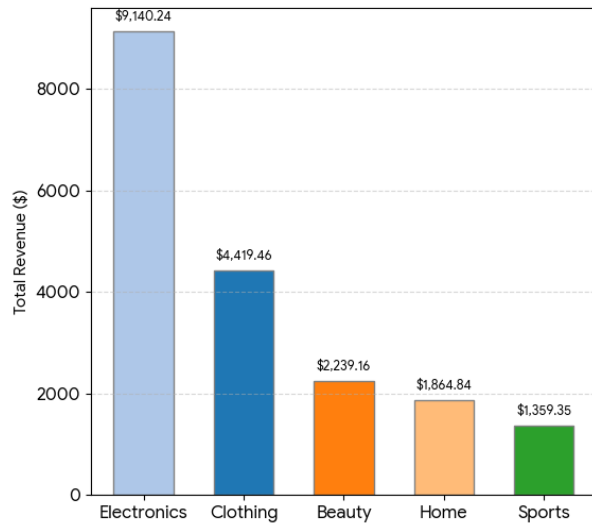
Pie Charts: Best for showing parts of a whole (keep categories under 5 for clarity).

Sparklines: These are tiny, single-cell charts perfect for showing trends over time next to data rows. Go to Insert > Sparklines (Line or Column) and select your data range.

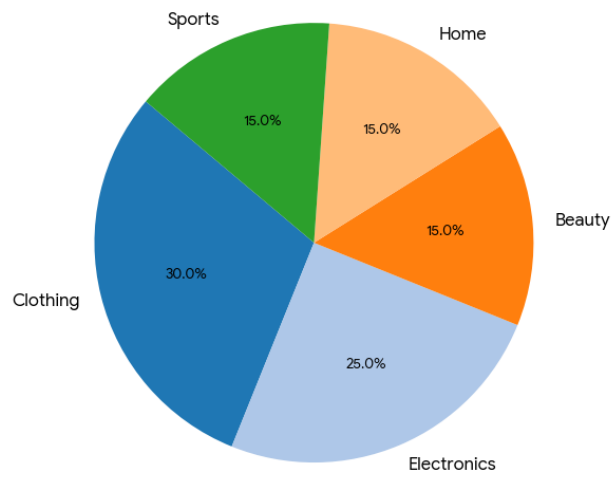
Step 3: Combine into a Dashboard

To make a basic dashboard, copy your charts onto a fresh, blank sheet. Go to the View tab and uncheck Gridlines to give it a clean, professional application look. Add Slicers (Insert > Slicer connected to your Pivot Table) so a user can click buttons to filter the charts instantly.

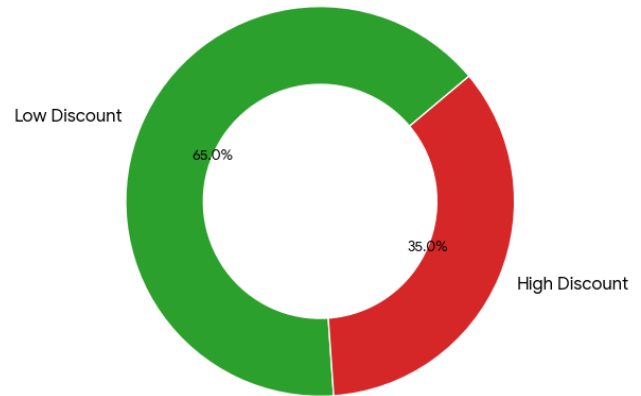
Total Transaction Value by Category (\$)



Distribution of Product Categories



Proportion of Discount Tiers Given



Distribution of Payment Methods

